

UPLOAD SLOT 01 OF 12 · DESIGN NARRATIVE &amp; CONCEPT

# Design Narrative & Concept

Falaj Al Safa — an arc of shade for a park that is currently too hot to use

Of the 4,402 daylight hours in a Dubai year, only **44.5%** are comfortable to stand in on the open site today. This proposal raises that to **64.6%** — a gain of 20.1 percentage points, or about 884 additional usable hours a year.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m<sup>2</sup> · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 03 August 2026 · every quantity regenerated by `python run_analysis.py`

## 1. The problem, stated as a number

A neighbourhood park nobody can stand in is not a park. Al Safa 2's difficulty is not layout, planting or maintenance — it is thermal. The site is flat and largely exposed, at 25°N, where the summer sun is nearly overhead and the winter sun is low enough to slide underneath most shade.

The analysis behind this submission reconstructs an 8,760-hour year for the site from 39 years of National Centre of Meteorology monthly normals, verified back against those normals to within 0.39 °C. Across the 4,402 daylight hours in that year, the open site is comfortable for 44.5% of them. That is the number the design exists to move.

## 2. The concept

**One arc.** A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

| Element   | What it is          | Description                                                                       |
|-----------|---------------------|-----------------------------------------------------------------------------------|
| Al Hilal  | the Crescent Canopy | 18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre               |
| Al Falaj  | the water channel   | 0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less |
| Al Nakhil | the Oasis Basin     | a sunken palm court held in the crescent's concave side                           |
| Al Sikkak | the alleys          | radial — each one a radius of the same arc                                        |
| Al Madar  | the perimeter loop  | a tree-shaded running and walking circuit                                         |
| Al Kathib | the dune berm       | planted earth against the roads — noise, glare and heat                           |

The six named elements. All are generated from a single arc definition in `src/plan.py`.

## 3. Why an arc — the part that was solved, not styled

A straight canopy presents one orientation. When a sun angle defeats it, it defeats the whole length at once, and the walk has no shade anywhere along it. An arc changes heading continuously, so some segment is always angled well.

The depth of the bow was not chosen for appearance. It was swept against the 8,760-hour solar model at a fixed section, over all 4,402 daylight hours of the year. The criterion was the final column — the hours in which the route offers nowhere at all to stand.

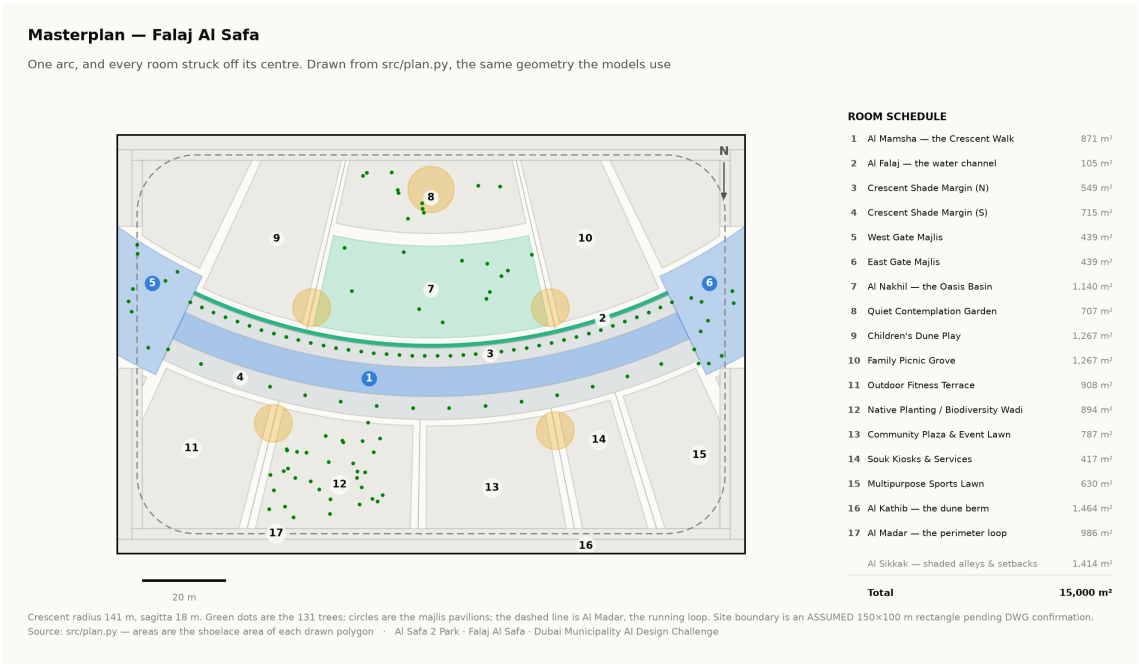
| Plan form                 | Mean cover | Worst month | Hours with no shade anywhere |
|---------------------------|------------|-------------|------------------------------|
| Straight east–west bar    | 87.4%      | 68.7%       | 330                          |
| Sine meander (superseded) | 85.0%      | 73.1%       | 63                           |
| Arc, sagitta 10 m         | 87.1%      | 70.2%       | 116                          |
| Arc, sagitta 14 m         | 86.6%      | 71.3%       | 62                           |
| Arc, sagitta 18 m         | 85.9%      | 72.1%       | 52                           |
| Arc, sagitta 22 m         | 84.9%      | 72.3%       | 61                           |
| Closed elliptical loop    | 79.1%      | 69.8%       | 89                           |

Adopted row highlighted. Source: the sweep recorded in src/config.py, measured against the same solar model used throughout this submission.

**Read the first column carefully.** The straight bar has the *highest* mean coverage of anything tested — an east–west canopy is close to the optimum orientation for 25°N, and curving it costs about a point. The crescent is not claimed to shade more ground on average. It is claimed to remove the hours in which the route offers nowhere at all to stand, and it removes six sevenths of them. That is the trade, stated in the direction that is not flattering.

The closed loop loses on every measure, for the same reason the deep bows do: it forces half its length to run north–south, and a canopy over a north–south route can only work when the sun is low in the east or west. The loop survives in the scheme as Al Madar, an *unshaded* running circuit — because that is what a circuit is actually for.

The arc bows convex south. The solar model is indifferent to that sign, so the choice was spatial: bowing south puts the structure between the sun and the hollow it wraps, making the concave side the park's cool pocket instead of a south-facing bowl. Every room people are asked to linger in sits in that hollow.



Masterplan. Every area is the shoelace area of the drawn polygon, which is why the schedule closes on 15,000 m<sup>2</sup> without anyone reconciling a spreadsheet. Technical drawing — to scale.

4. What the design achieves

| Measure                                  | Today   | As designed |
|------------------------------------------|---------|-------------|
| Comfortable daylight hours               | 44.5%   | 64.6%       |
| Peak heat index                          | 56.8 °C | 48.7 °C     |
| Mean heat-index reduction under canopy   | —       | 7.13 °C     |
| Crescent Walk shaded (canopy and louvre) | —       | 87.3%       |
| Site-wide mean shade                     | —       | 34.1%       |
| Trees                                    | —       | 131         |

**Site-wide mean shade is modest at 34.1%, and that is a position, not an oversight.** This scheme makes a few places genuinely excellent rather than the whole site marginally better. A park with even shade everywhere and no continuously comfortable route would score higher on that one statistic and be worse to use.

## 5. Two corrections this submission makes to itself

**A withdrawn shade claim.** An earlier version claimed 99.2% annual shade on a flat 9 m canopy over a 9 m walkway. It does not survive a geometric check — when the sun is low and to the south, the shadow slides clean off the path — and it was withdrawn. The section was then re-solved: a narrower walk, a wider overhang, a lower plane and a deep southern louvre. It now measures 87.3%, and that is a number that can be checked.

**Three withdrawn images.** Three visuals presented invented data as measurement: a thermal comfort analysis attributed to CFD when no CFD was run; a satellite NDVI analysis whose raster was noise; and a parametric canopy described as solar-optimised when no objective was ever defined. All three were withdrawn on the same principle as the 99.2%: anything that cannot survive being checked should not be in front of a juror.